

TrES-1b

R-1289

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-1_210503 · created 2026-07-31T16:00:49+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2459337.8462 +/- 0.0014 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1555 +/- 0.007
Transit depth (Rp/Rs)^2	2.42 +/- 0.22 [%]
Orbital Inclination (inc)	89.97 +/- 0.57
Airmass coefficient 1 (a1)	1.211 +/- 0.012
Airmass coefficient 2 (a2)	-0.168 +/- 0.0027
Scatter in the residuals of the lightcurve fit is	0.96 %
Best Comparison Star	None
Optimal Aperture	4.24
Optimal Annulus	17.49
Transit Duration (day)	0.10569 +/- 0.00088

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.1555 ± 0.007 vs 0.1358 (deviation 2.81σ)
Duration fit vs published	0.10569 d vs 0.1042 d (deviation 1.73σ)
Mid-time O–C	-3.9 min
Depth SNR	22.2
Residual scatter	0.96 %

Light curve

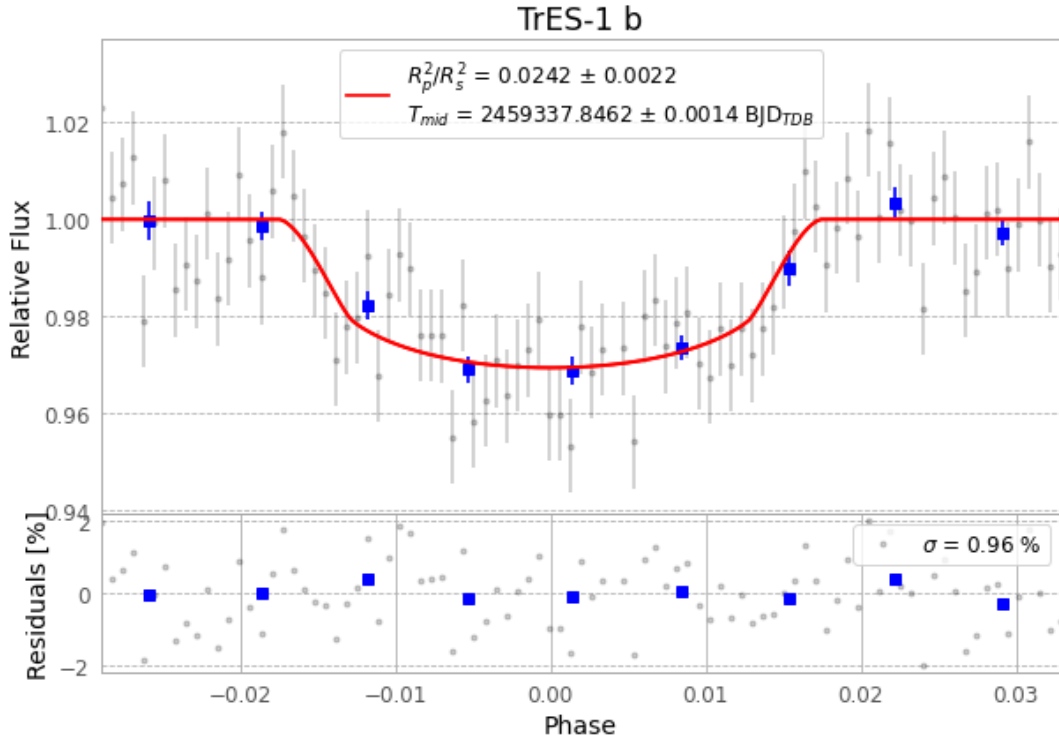


Figure 1 — FinalLightCurve_TrES-1 b_2021-05-02.png (SHA-256 1cedd24fca31ef37...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [280, 257], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 75c81bb647eb90d8...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

91 FITS frames (60 MB), TRES-1210503060912.FITS → TRES-1210503103912.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-1-210503.log (3bdeddbb1dea...), FinalLightCurve_TrES-1 b_2021-05-02.png (1cedd24fca31...), FinalLightCurve_TrES-1 b_2021-05-02.csv (defb9bc85fb7...)

Compute resources

Wall time 150.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 16:00Z.